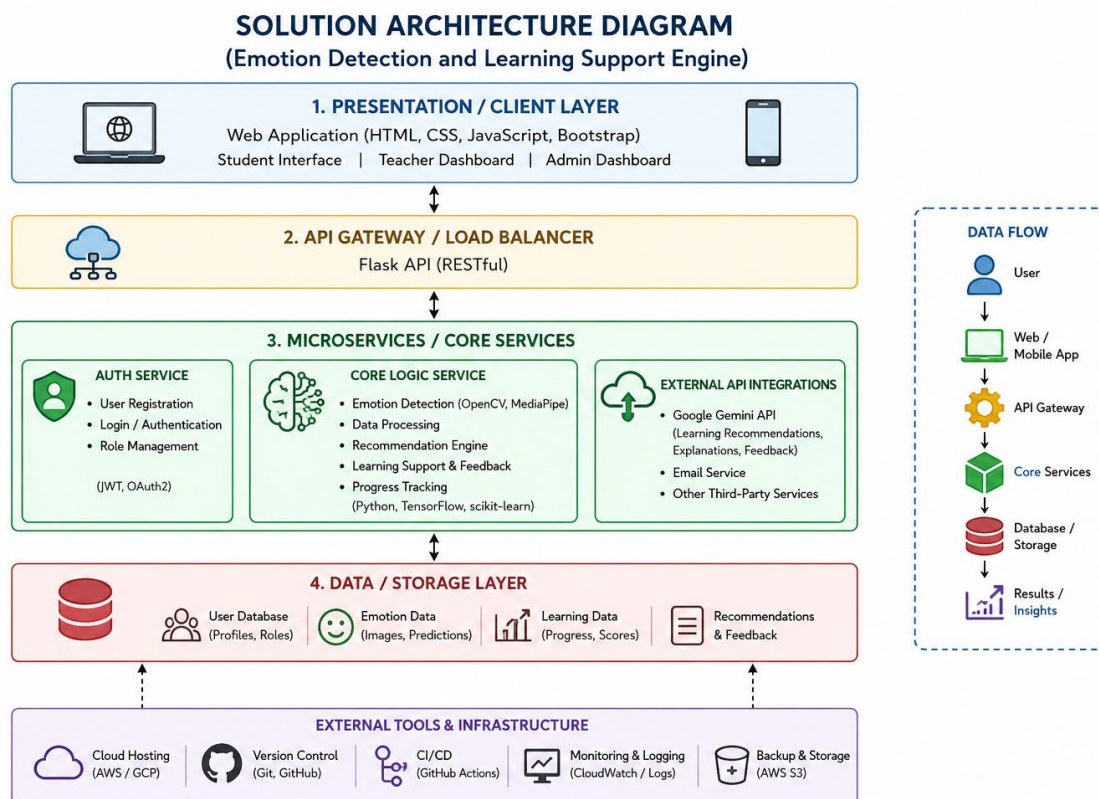


Solution Architecture

Date	27 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface for students and teachers to interact with the system.</i>	<i>HTML, CSS, JavaScript, Bootstrap</i>
<i>API Gateway</i>	<i>Handles client requests and routes them to backend services.</i>	<i>Flask</i>
<i>Microservices</i>	<i>Performs authentication, emotion detection, recommendation generation, and progress tracking.</i>	<i>Python, TensorFlow, OpenCV, Media Pipe, Gemini API</i>
<i>Database</i>	<i>Stores user profiles, detected emotions, learning history, and recommendations.</i>	<i>SQLite (or MySQL)</i>